Week 1 Internship Progress Report

Python Programming & Data Analysis Internship

Presented By: Chris | **Company:** Srishti Innovatives | **Duration:** Week 1

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Internship Overview

The first week of the internship was designed to build a strong technical foundation across Python programming, data manipulation, and visualization — skills directly applicable to real-world software and analytics environments. Each day introduced progressively more complex concepts, culminating in hands-on projects and dataset analysis.

Technologies Learned

- Python — core programming language
- NumPy — numerical computing
- Pandas — data manipulation
- Matplotlib & Seaborn — visualization
- GitHub — version control

Main Areas Covered

01

Python Fundamentals

Data types, operations, and syntax

02

Mini Projects

Hangman game, file-based user system

03

Data Analysis

NumPy arrays, Pandas DataFrames

04

Visualization

Charts, plots, and heatmaps

05

Version Control

GitHub repository management

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DAY 1

Company Induction & Orientation

The first day set the tone for the entire internship. Orientation activities focused on aligning expectations, understanding company culture, and establishing the professional standards required throughout the program. This session also introduced the technical roadmap and the domains that would be covered in the coming weeks.

Company Introduction

Overview of Srishti Innovatives, its mission, and the team structure interns would be working within.

Guidelines & Ethics

Clear Do's and Don'ts were outlined, covering professional conduct, communication norms, and work expectations.

Workflow Overview

Introduction to the internship workflow, task submission process, and how progress would be tracked weekly.

Technical Domains

A preview of the courses and technical areas — Python, data analysis, and visualization — that would be explored.

Key Learning Outcome: Understanding professional workplace culture and the importance of technical skills in industry environments.

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DAY 2

Python Fundamentals

Day 2 was dedicated to mastering Python's core data structures and operations. Through a combination of guided exercises and independent practice, all foundational syntax concepts were covered and reinforced with hands-on problem-solving.

Topics Learned

String Operations

Manipulation, formatting, and built-in methods

List & Tuple Operations

Indexing, slicing, and iteration

Dictionary & Set Operations

Key-value pairs and set theory

Practical Work

- Solved multiple Python practice questions covering all data types
- Successfully executed all exercises with correct output
- Developed a fully functional **Hangman Game** using Python — applying conditionals, loops, and user input handling

Skills Improved

- Logical Thinking
- Problem Solving
- Python Syntax Understanding

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DAY 3

File Handling, Exception Handling & Mini Project

Day 3 introduced file I/O operations and error management — critical skills for building robust, production-ready Python applications. These concepts were immediately applied in a structured mini project that combined all learned topics into a cohesive system.

Topics Covered

→ File Reading & Writing

Opening, reading, writing, and closing files using Python's file handling methods

→ Exception Handling

Using `try-except` blocks to gracefully manage runtime errors

→ Module Importing

Importing built-in and custom modules to organize code efficiently

🗂️ Mini Project: Smart File-Based User Management System

A modular Python application that demonstrates real-world file handling and data persistence.

Store Users

Persist user information to file

Read & Update

Retrieve and modify existing records

Error Handling

Graceful failure on invalid input

Modular Code

Organized, reusable functions

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DAY 4

NumPy, Pandas & Data Visualization

Day 4 marked the transition from core Python to data-focused libraries. NumPy and Pandas were introduced as the backbone of data analysis workflows, while Matplotlib and Seaborn enabled visual storytelling from raw data.

NumPy Arrays

Created and manipulated multi-dimensional arrays, performed mathematical operations, and explored array indexing and broadcasting.

Pandas DataFrames

Loaded datasets, explored DataFrame structures, applied filtering, grouping, and data cleaning techniques to prepare data for analysis.

Statistical Analysis

Calculated descriptive statistics including mean, median, standard deviation, and correlation matrices to understand data distributions.

Visualization Tools

Practiced creating Bar Graphs, Histograms, Scatter Plots, and Heatmaps using Matplotlib and Seaborn to communicate insights visually.

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DAY 5

Dataset Analysis Project — Titanic & Iris

The final day of Week 1 was dedicated to applying the full EDA (Exploratory Data Analysis) workflow on real-world datasets. The Titanic dataset was provided as the primary task, and the Iris dataset was independently selected for additional practice.

📊 Titanic Dataset

Analyzed passenger survival data to uncover patterns related to age, class, and gender — a classic EDA exercise in data science.

- Data cleaning and missing value handling
- Survival rate analysis by category
- Correlation heatmaps and visual summaries

🌸 Iris Dataset (iris.csv)

Chosen as a supplementary Kaggle dataset to independently practice the full EDA pipeline on a well-known classification dataset.

Data Cleaning

Checked for nulls and duplicates

Correlation Analysis

Explored feature relationships

Visualizations

Heatmaps, histograms, pairplots, scatterplots

Learning Outcome: Gained hands-on experience with the complete Exploratory Data Analysis (EDA) workflow — from data ingestion to insight generation.

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GitHub Repository Submission

All weekly deliverables — including Python exercises, mini projects, and dataset analysis notebooks — were organized and uploaded to GitHub, demonstrating version control proficiency and professional code submission practices.

🐍 Python Practice Programs

All string, list, tuple, dictionary, and set operation exercises completed during Day 2.

🎮 Hangman Game

Fully functional Python game applying conditionals, loops, and user input handling.

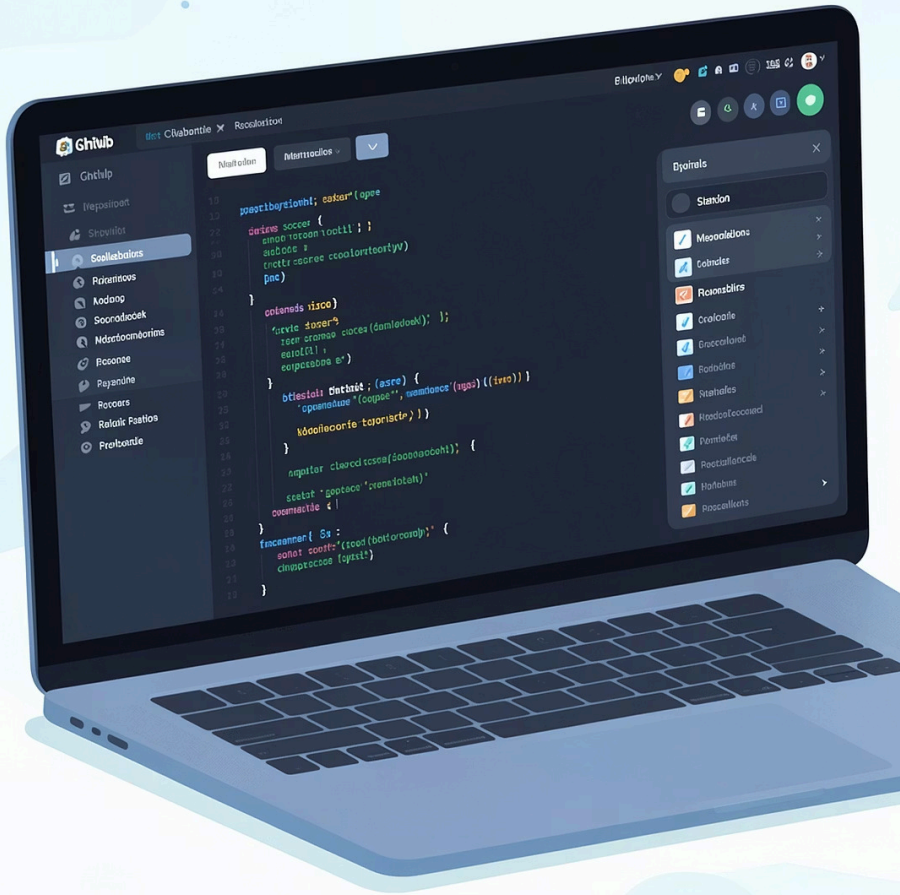
🗂️ User Management Project

File-based system with read, write, update, and exception handling capabilities.

📈 Data Analysis Files

NumPy & Pandas tasks, Titanic dataset analysis, and Iris dataset EDA notebooks.

Repository: [2-week-Srishti-Innovatives-Internship Repository] — All Week 1 work is publicly accessible and version-controlled.



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Skills Gained This Week

Week 1 delivered a well-rounded set of technical and professional competencies. The combination of guided learning, independent projects, and real dataset analysis accelerated skill development across multiple domains.

Technical Skills

Python Programming

Core syntax, data structures, file I/O, and modular code design

Data Analysis

NumPy arrays, Pandas DataFrames, and statistical operations

Data Visualization

Matplotlib and Seaborn for charts, plots, and heatmaps

GitHub Version Control

Repository management, commits, and code submission

Soft Skills

Problem Solving

Breaking down complex tasks into manageable steps

Logical Thinking

Structuring code and algorithms with clear reasoning

Professional Communication

Adapting to workplace norms and reporting progress clearly

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Challenges Faced & Solutions

Every learning curve comes with obstacles. This week presented a few technical challenges that were systematically addressed through practice, debugging, and peer-supported learning — turning each difficulty into a growth opportunity.

⚠️ Challenges Encountered

- **Debugging Python errors** — Initial difficulty identifying and resolving syntax and runtime errors independently
- **Handling dataset file paths** — Issues with relative vs. absolute paths when loading CSV files
- **Understanding visualization libraries** — Matplotlib and Seaborn syntax required time to internalize

✅ Solutions Applied

1 Continuous Practice

Repetitive hands-on coding reinforced concepts and reduced error frequency over time

2 Debugging Techniques

Used print debugging, error message analysis, and step-by-step code review to isolate issues

3 Implementation-Based Learning

Learning by doing — building projects and visualizations directly accelerated library familiarity

Week 1 Summary: A strong foundation has been established across Python, data analysis, and visualization. Ready to build on these skills in Week 2 with more advanced projects and deeper dataset exploration.

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